HASSAN TAHIR

STRUCTURE ENGINEER/PROJECTS COORDINATOR

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PROFESSIONAL SUMMARY

Structural Engineer with 5 years of professional experience delivering structural design, seismic evaluation, and retrofit solutions for buildings, industrial facilities, bridges, and specialized structures. Strong background in elastic and performance-based seismic analysis, including response spectrum, time-history, and pushover methods. Experienced in applying international codes and standards such as *American and EN 1990–1993*. Skilled in coordinating with Infrastructure, Quantity Estimation, Architecture, and MEP departments to ensure integrated and efficient design development. Demonstrated success in managing projects from conception to construction support, delivering resilient and cost-effective designs that comply with international standards.

PROFESSIONAL EXPERIENCE

DESIGNMEN CONSULTING ENGINEERS, ISLAMABAD **Sept 2021 - Mar 2024, Dec 2024 - Present**

HEAD OF STRUCTURAL DEPARTMENT & PROJECTS COORDINATOR

Lead structural design and coordination for building and industrial projects, overseeing analysis, assessment, and retrofitting works. Perform structural design and manage a team of engineers, ensuring technical accuracy, client coordination, and timely project delivery.

Responsibilities

- Utilized advanced Finite Element Analysis (FEA) software to develop and optimize structural models, ensuring efficient and cost-effective designs that meet or exceed client expectations.
- Prepare technical reports and documentation outlining design concepts, calculations, and recommendations.
- Effective coordination with clients including monthly and fortnightly meetings, and site staff to gather site progress information.
- Performed construction support services and communicated effectively with contractors during RFI's, shop drawing reviews, and structural observations.
- Top supervision of projects throughout the construction phase.
- Coordination with lead architect, and MEP engineers throughout the project.

GULF GRC, DOHA, QATAR **Apr 2024 - Dec 2024**

STRUCTURE DESIGN ENGINEER/PROJECTS COORDINATOR

Delivering innovative structural engineering solutions for facade systems and steel structures, including warehouses.

Responsibilities

- Analyze and design structural systems, including facade elements, as well as steel structures such as warehouses and building frameworks, ensuring compliance with BS EN and American standards.
- Conduct advanced steel connection design and structural analysis using Robot Structural Analysis, IdeaStatica, ANSYS, ABAQUS, and StaadPro.

PROFESSIONAL CONTRIBUTIONS

- Authored a guide for Pakistani structural engineers transitioning to the new Building Code of Pakistan 2021.
- Successfully advocated for the adoption of Building Information Modeling (BIM) within the structural design department.
- Achieved a significant outcome for the company, contributing to its financial success and growth by successfully obtaining price adjustments from the client in the face of a year-long country-wide price hike.
- Facilitated and conducted office-wide lectures on Building Information Modeling (BIM) and instructed software tools like Revit, etc., for the drafting (CAD) section, enhancing team skills and project efficiency.
- Simultaneously managed responsibilities across three departments: Structure, Project Coordination, and Bid Coordination, demonstrating adaptability and multitasking capabilities.
- I have also mentored summer internship trainees from various universities, providing comprehensive training in various civil engineering software, enhancing their skillsets and professional growth.

SOFTWARE SKILLS

Building Codes: ACI 318, ASCE 7-16, ASCE 41-17/23, AISC, BS EN, UBC-97, FEMA,356/P-154, ATC 40
Concrete Structure Analysis and Design: Etabs, Sap 2000, Safe, Staad Pro, ASDIP Foundation
Steel & Connection Design: Robot Structural Analysis, Sap 2000, Staad Pro, Idea Statica, ASDIP Steel
Performance-Based Design: CSI Perform 3D
BIM and Documentation: Revit, Oracle Aconex, Revu Bluebeam, AutoCAD
Manual Calculation Tools: Mathcad, Tekla Tedds
Soft Skills: Team player, Attention to detail, Problem-solving, Bias for action, Deliver results

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EDUCATION

Masters of Science in Structural Engineering **2026**
University of Engineering and Technology, Taxila
Bachelors of Science in Civil Engineering **2021**
Comsats University Islamabad, Wah Campus

CERTIFICATIONS

- Construction Project Management | (Columbia University)
- Foundation of Structural Dynamics | (ISAE-SUPAERO - Toulouse University)
- Profis Engineering Suite x Idea Statica | (XStructures)
- Design of Steel Warehouse on Robot Structural Analysis | (Civil Engineering Essentials)
- Etabs Professional Course | (Engineering Learning Platform)
- SAP2000 Professional Course | (Engineering Learning Platform)
- Space Frame: Theory, Structural Analysis & Design by ETABS | (Akshay Thakur)
- CSI Safe + All Pile foundation, slabs, beams and deflection | (Ahmad Traboulsi)
- BIM- Revit Structure Full Course- from Beginner to Advanced | (Ghasem Ariyani)
- BIM- Revit Steel Structure Project- From Zero to Advanced | (Ghasem Ariyani)

KEY PROFESSIONAL PROJECTS

Structural Design & Detailed Engineering (Buildings and Industrial Structures)

- Structural Design and Seismic Performance Evaluation of a 13-Story RC Dual Wall–Frame Residential Apartment Building (HARSUKH-2), Ayubia.
- Nonlinear Seismic Response Evaluation of an 11-Story HHRD Building in Islamabad using Response Spectrum and Time History Analysis.
- Code-Based Seismic Performance Evaluation of a 06-Story RC NUML University Building, Faisalabad, using Response Spectrum Analysis.
- Structural Design of Skilled Development Institute and Medical & Diagnostic Center, Qurtaba City.
- Structural Design of Auditorium Building and Islamic Center (AAGHOSH Project).
- Structural Design of Higher Education Commission (HEC) Secretariat Building, Islamabad (NESPAK Design – Independent Vetting).
- Structural design of 03-story residential apartment buildings (2-bed and 3-bed units) in Aiwan-e-Saddar, Islamabad.

Performance-Based Seismic Evaluation & Earthquake Engineering

- Performance-Based Seismic Evaluation of a 09-Story RC Building Using ASCE 7-16 AND ASCE 41-17 (Frequency Allocation Board), Islamabad.
- Site-Specific Deterministic Seismic Hazard Assessment (DSHA) for a Hydropower Project in the Himalayan Fold-and-Thrust Belt.
- Performance-Based Structural Evaluation of CDA Hospital Buildings (Blocks 1–6), G-6 Islamabad using pushover analysis.
- *ASCE 41-23* Tier 1 and Tier 2 Seismic Evaluation of CABI Regional Bioscience Center, Rawalpindi.
- Rapid Visual Seismic Screening of CDA Satellite Clinics in Islamabad *(FEMA P-154)*.

Structural Assessment, Retrofitting & Rehabilitation

- Structural Assessment and Retrofitting Design of Multi-Block Educational Buildings, including foundation strengthening and vertical expansion.
- Structural Assessment and Retrofitting of a 2-Story Hospital Building in Karachi, achieving 57% drift reduction through shear wall addition and section enhancement.
- Comprehensive Structural Evaluation and Rehabilitation Design of an Existing Bridge, including embankment stabilization, lane addition, and bridge extension.

Industrial, Warehouse & Crane-Supported Structures

- Engineering Design of a Large-Scale Steel Warehouse Integrated with a 10-Ton Overhead Crane *(Eurocodes EN 1990/1991/1993).*
- Design of a Mid-Scale Steel Warehouse with 5-Ton Crane using Eurocodes and Robot Structural Analysis.
- Structural Design of Two PMO Office Warehouses (4,000 sq ft each) accommodating 5-Ton Cranes using SAP2000, ASDIP Steel, and IDEA StatiCa.
- Structural Design of Steel Winch System for BMU Roof Operations (800 kg panels) using *BS EN 1993, EN 1808, and AISC LRFD.*

Design Coordination & Construction Support

- Engineering Design Coordination for 12 School Projects for Aga Khan Education Services (Rs. 1.857 Billion portfolio) in Gilgit-Baltistan and Chitral.
- Construction Phase Design Support and Supervision across multi-disciplinary project teams.

Special Structures & Non-Conventional Systems

- Dynamic Structural Response Evaluation of an Elevated Water Tank under Blast-Type Transient Loading at PMO.
- Design of 12 Steel Structures for Escape Hotel Projects at Nathia Gali, including a 15-ft cantilevered floating restaurant (AISC, ASCE).
- Structural Design of Iconic Lake Downtown Fountain Structures (foundations, steel floor system, and 6×28 ft bridge span).

Façade, GRC & Panelized Systems Engineering

- Design Coordination and Structural Analysis of Composite GRC Wall Panels, including thermal, dew point, and acoustic assessments using ANSYS.
- Analysis and Design of GRC Elements (columns, cornices, brackets) for Four Palaces Project (IB+G+1), Al Wajba, Qatar.
- Structural Analysis and Design of GRC Perforated Panels for Water Sports Centre, Cluster 6, Sindalah Island, NEOM, Saudi Arabia (SBC 301, AISC LRFD).

PROFESSIONAL WRITINGS

- Enhancing Structural Integrity and Resilience: A Systematic Approach for Retrofitting Design of a Two-Story Hospital Building in Karachi. *(CSCE'24/1st Author)*
- Enhancing Structural Design Efficiency in Pakistan: A Case Study on The Impact of Bim Integration.*(CSCE'24/1st Author)*
- Enhancing Concrete Mechanical Properties Through Graphite Nano-Micro Platelets. *(CSCE'24/1st Author)*
- Efficiency of Water Supply System of Kaghan Valley during High Influx of Visitors. *(ICACEE-2024)*

AWARDS AND APPRECIATIONS

- Received an Appreciation Letter from Aga Khan Education Services, Pakistan.
- Received an Appreciation Letter from Iqra University Islamabad, Pakistan.
- Received an Appreciation Letter from Helping Hand Organization.